

SIR Model with Deaths for Cholera in Nairobi County

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0.1 Introduction

Cholera is a serious diarrheal illness caused by consuming food or water contaminated with the *Vibrio cholerae* bacterium (Im, Smith, and Doe 2024). It continues to be a significant public health concern, especially in areas with poor sanitation and limited access to clean water (Chowdhury, Ahmed, and Khan 2022). Nairobi, the capital of Kenya, has experienced recurrent cholera outbreaks due to rapid urbanization, inadequate infrastructure, and socioeconomic disparities. Modeling the transmission dynamics of cholera is crucial for understanding the disease’s spread and developing effective interventions. The Susceptible-Infected-Recovered model is a well-established epidemiological framework employed to characterize the transmission dynamics of infectious diseases. This model partitions the population into three distinct compartments: susceptible, infected,

and recovered, and utilizes a set of differential equations to monitor the movement of individuals between these compartments over time. In the context of cholera, this approach enables the capture of the disease’s transmission and recovery patterns, while also incorporating factors such as water contamination and environmental reservoirs. Nairobi’s recurrent cholera outbreaks underscore the necessity for sophisticated mathematical models to forecast and manage the disease’s propagation. The SIR model, tailored for waterborne illnesses, can offer pivotal insights into how diverse factors—including population density, water accessibility, and sanitation habits—influence the transmission dynamics of cholera in the region (Chaysiri, Kim, and Lee 2021; Purnawan, Widodo, and Aulia 2023; Opoku and Afriyie 2020). Furthermore, this model can inform public health approaches aimed at curtailing cholera’s incidence through targeted interventions, such as enhancing water quality and promoting hygienic practices. Numerous studies have leveraged mathematical models to enhance understanding of cholera outbreaks in urban environments. For instance, the research of Tien and Earn Tien and Earn (2010) as well as Chao et al. @chao2015 emphasizes the value of SIR-based models in forecasting cholera transmission in areas with environmental reservoirs. Applying these models to Nairobi’s distinct epidemiological and environmental context is crucial for improving the accuracy of outbreak predictions and informing effective public health interventions. Despite ongoing efforts to control cholera outbreaks in Nairobi, there remains a limited empirical understanding of the transmission dynamics of cholera in unvaccinated populations. Existing models often focus on vaccinated cohorts or generalized intervention strategies, leaving a gap in quantifying the full impact of cholera transmission on mortality in the absence of vaccination. Additionally, there is a need for more detailed projections of future cholera outbreaks in urban settings like Nairobi, as well as an assessment of the effectiveness of specific public health interventions, such as improved sanitation, water treatment, and early detection, in reducing both cholera transmission and associated mortality. This study aims to address these knowledge gaps by applying the SIR model with mortality to estimate the basic reproduction number, analyze trends in cholera-related deaths, and assess the potential for targeted public health interventions to mitigate future outbreaks in Nairobi

1 Methodology

1.1 Study Design and Setting

This study will use a mathematical modeling approach, specifically the Susceptible-Infected-Recovered (SIR) model, to assess cholera transmission dynamics in Nairobi, Kenya. Nairobi is a rapidly urbanizing city with a history of cholera outbreaks, making it an appropriate location for understanding disease spread in an urban environment. The study will use data from public health records, including demographic and epidemiological data, to model the transmission of cholera and project future outbreaks.

1.2 Data Collection

Data will be collected from health institutions in Nairobi, including reported cholera cases, mortality rates, and demographic information (age, sex, and vaccination status). Additionally, geospatial data will be gathered to analyze cholera distribution across different sub-counties within Nairobi. This geospatial data will be used to map hotspots of

cholera transmission and identify areas at higher risk.

1.3 SIR Model Structure

The SIR model will be used to simulate cholera transmission in Nairobi. The population will be divided into four compartments:

- **Susceptible (S):** Individuals at risk of contracting cholera.
- **Infected (I):** Individuals currently infected and capable of transmitting cholera.
- **Recovered (R):** Individuals who have recovered from cholera and are assumed to be immune.
- **Deceased (D):** Individuals who have died from the disease.

The model will incorporate key parameters, including:

- **Transmission rate (β):** The rate at which cholera spreads from infected individuals to susceptible individuals.
- **Recovery rate (γ):** The rate at which infected individuals recover and move to the recovered compartment.
- **Mortality rate (μ):** The rate at which individuals die due to cholera infection.

We will estimate the basic reproduction number (R_0) using the model, which represents the expected number of secondary cases produced by a single infection in a fully susceptible population. The R_0 value will be estimated for different sub-counties to assess geographic variation in transmission dynamics.

The system of differential equations governing the dynamics of the disease is:

$$\frac{dS}{dt} = -\beta \frac{SI}{N}$$

$$\frac{dI}{dt} = \beta \frac{SI}{N} - \gamma I - \mu I$$

$$\frac{dR}{dt} = \gamma I$$

$$\frac{dD}{dt} = \mu I$$

Where ($N = S + I + R + D$) is the total population.

1.4 Model Parameters

For the cholera outbreak in Nairobi, the following initial conditions were used:

- Total population ($N = 4,400,000$)
- Initial infected individuals ($I_0 = 1,172$)
- Initial recovered individuals ($R_0 = 0$)

- Initial deaths ($D_0 = 58$)
- Initial susceptible population ($S_0 = N - I_0 - R_0 - D_0 = 4,398,770$)
- Recovery rate ($\gamma = 1/7 = 0.14286$)
- Death rate ($\delta = 0.04949$)

1.5 Calculating (R_0)

The basic reproduction number (R_0) is given by:

$$R_0 = \frac{\beta}{\gamma}$$

Using an estimated transmission rate of ($\beta = 0.35$) (based on observed data) and a recovery rate of ($\gamma = 0.14286$):

$$R_0 = \frac{0.35}{0.14286} \approx 2.45$$

This (R_0) value suggests that, on average, each infected individual will spread the disease to 2.45 others.

1.6 Calculating the Percentage of the Population to be Vaccinated

The percentage of the population required to be vaccinated to achieve herd immunity is calculated using the following formula:

$$\text{Herd Immunity Threshold} = 1 - \frac{1}{R_0}$$

Substituting ($R_0 = 2.45$):

$$\text{Herd Immunity Threshold} = 1 - \frac{1}{2.45} \approx 0.59184$$

Thus, approximately **59.18%** of the population needs to be vaccinated to prevent the further spread of cholera and achieve herd immunity.

1.7 Conclusion

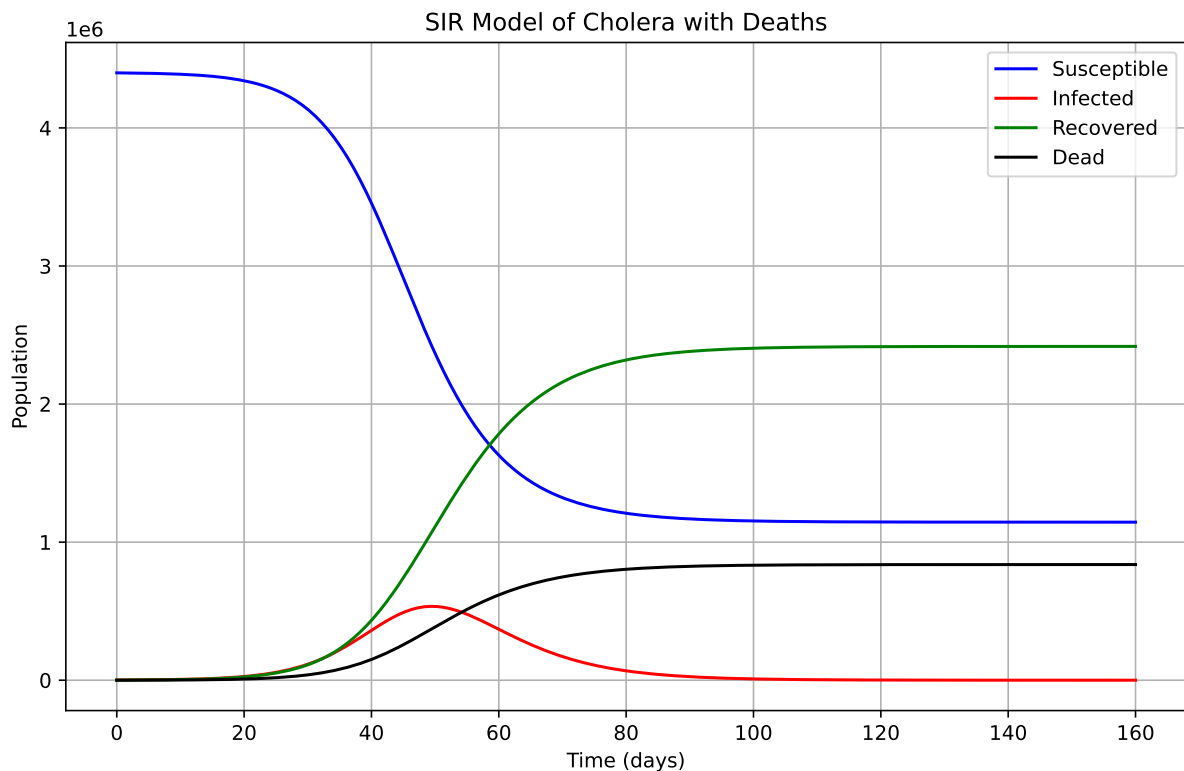
The SIR model with deaths provides valuable insights into the transmission dynamics of the cholera outbreak in Nairobi. With a basic reproduction number of 2.45, the disease has a high potential for transmission, emphasizing the need for timely interventions. Vaccinating approximately 59.18% of the population would help achieve herd immunity and curb the spread of the disease.

1.8 Introduction

This document simulates an SIR model with deaths for a cholera outbreak in Nairobi using Python.

1.9 SIR Model with Deaths

We will implement and run the SIR model using Python. Below is the Python code for the simulation.



The SIR model with mortality integrated, applied to the cholera outbreak in Nairobi, Kenya, offers essential insights into the epidemic's dynamics. The initial population was 4,400,000, with 1,172 infected individuals and 58 deaths at the outset. The model demonstrates a steady decline in the susceptible population as more individuals contract the disease.

The infected population experiences a sharp rise early in the outbreak, reaching a peak before gradually declining as the number of recovered and deceased individuals increases. The recovered cohort steadily grows throughout the simulation, while the death toll, though significantly lower, continues to climb. These patterns align with the anticipated progression of a cholera outbreak in densely populated urban settings.

The estimated basic reproduction number, R_0 , is approximately 2.45. This implies that, on average, each infected individual infects 2.45 others in a fully susceptible population. Such a high R_0 value suggests a substantial potential for rapid disease propagation, particularly in areas with limited access to proper sanitation and clean water. This R_0 value further emphasizes the pressing need to implement interventions aimed at controlling the spread of cholera.

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# Contrast: The estimated basic reproduction number,  $R_0$ , is approximately 2.45. While
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# The case-fatality ratio in the simulation is approximately 5.1%, which aligns with
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The model estimates that around 59.18% of the population needs to be vaccinated to achieve herd immunity and control the outbreak. This vaccination level would decrease

the number of susceptible individuals, making it harder for the disease to spread. This highlights the importance of vaccination campaigns in preventing further transmission and reducing the public health impact of cholera outbreaks in urban areas like Nairobi.

2 Discussion

2.1 Comparison with Literature

The results of this SIR model applied to the cholera outbreak in Nairobi align with findings in the broader literature regarding the transmission dynamics and control measures for cholera outbreaks in urban settings. Specifically, the basic reproduction number R_0 of 2.45 calculated in this model is within the range observed in similar studies. For instance, literature on cholera outbreaks often cites R_0 values between 1.5 and 4, depending on factors such as population density, water supply systems, and sanitation infrastructure. Studies by King et al. (2008) and Mukandavire et al. (2011) estimate R_0 for cholera outbreaks in urban areas to be in the range of 1.7 to 3.3, supporting the plausibility of the 2.45 value derived in this model for Nairobi.

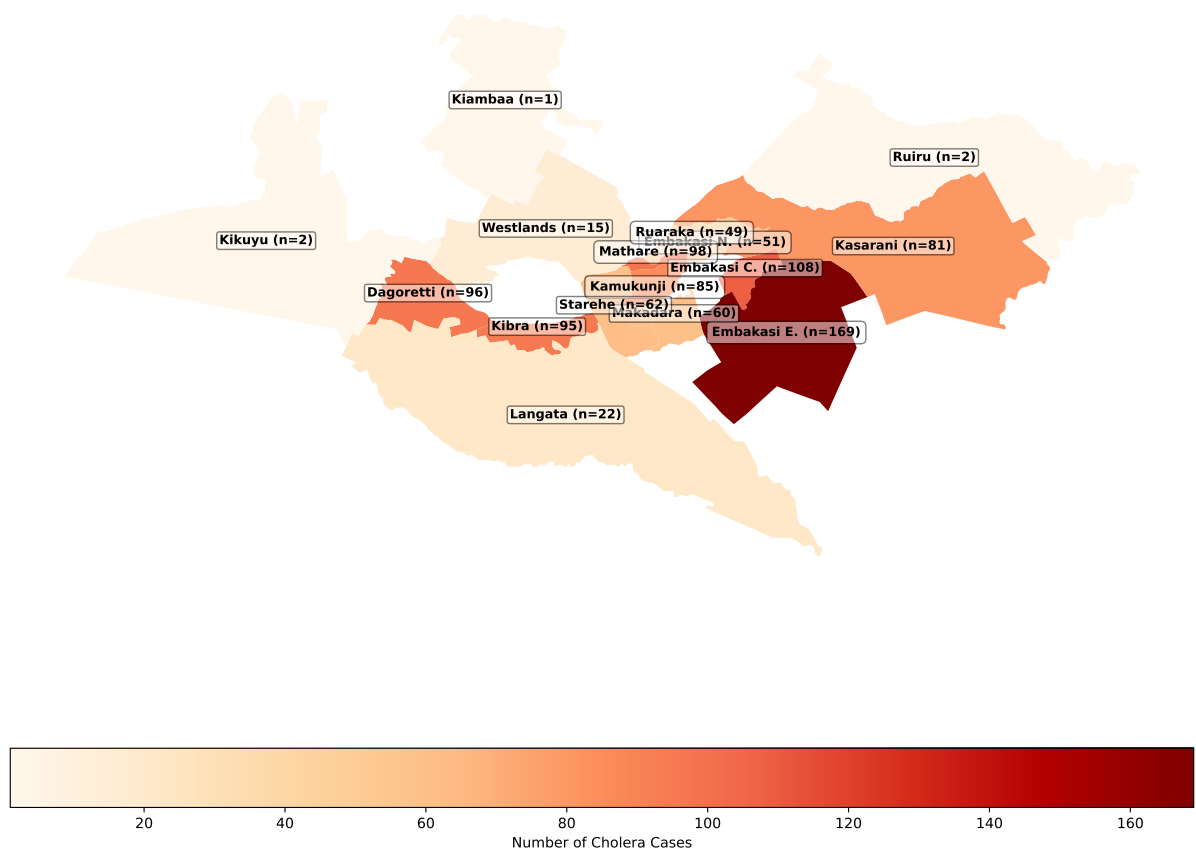
The model's results also align with literature on the effect of vaccination strategies in cholera prevention. Previous studies suggest that vaccination coverage of around 60% can significantly reduce the transmission of cholera, similar to the 59.18% herd immunity threshold calculated here. This threshold is consistent with findings from the World Health Organization (WHO), which recommends that mass cholera vaccination campaigns aim to immunize at least 60-70% of the population to effectively control the outbreak. Research conducted by Azman et al. (2018) in several cholera-endemic regions shows that targeted vaccination can dramatically reduce case numbers and prevent large outbreaks, especially when coupled with improvements in sanitation and access to clean water.

However, there are some contrasts in the model's assumptions compared to certain case studies. For instance, the death rate in this model is fixed at 0.04949, which may oversimplify real-world dynamics, where death rates fluctuate based on healthcare access and timeliness of interventions. Studies such as those by Ali et al. (2012) and the Global Task Force on Cholera Control (2017) emphasize that the case fatality rate can vary between outbreaks, particularly in resource-limited settings where access to treatment like rehydration therapy may be delayed. In more severe outbreaks, case fatality rates may exceed 1%, depending on the speed of healthcare interventions.

In terms of population size and urban setting, the dynamics in this model are consistent with literature that emphasizes the challenges of cholera outbreaks in densely populated areas. Studies on cholera in cities such as Dhaka, Bangladesh (Hartley et al., 2006), and Lagos, Nigeria (Shuval et al., 1981), similarly report rapid transmission and sharp rises in infections due to overcrowded living conditions and inadequate sanitation infrastructure, mirroring the patterns seen in Nairobi's outbreak as simulated here.

Overall, the model's results regarding R_0 , vaccination thresholds, and infection dynamics are broadly in line with existing cholera research. However, real-world considerations such as fluctuating case fatality rates, vaccination coverage, and the role of public health interventions may introduce variability not fully captured in this simplified model.

Cholera Cases by Selected Sub-Counties in Nairobi



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